

R49. What is FIFO (First-In, First-Out) Queuing, and how does it work?

- Packets wait in buffer until the node is ready to process them.
- If average arrival rate is higher than average processing rate, then queue will fill up and new packets will be discarded.
- FIFO has no concept of priority or classes of traffic
- There is only one queue, and all packets are treated equally.
- Packets are sent out an interface in the order in which they arrive.

Drawbacks of FIFO in a Congested Network:

- No Importance: FIFO does not account for packet importance.
- Delay: High-priority traffic gets delayed behind lower-priority traffic.
- Latency: Causes high latency (delays) for real-time applications like VoIP or Video Conferencing.

Scenarios of FIFO

- Works well in low-traffic networks.
- Suitable for networks that do not require strict QoS controls (e.g., internal systems where prioritization is not necessary)

R50. What is Priority Queuing (PQ), and how does it work? What are the advantages of using Priority Queuing? What are the disadvantages of Priority Queuing? Give an example where Priority Queuing would be beneficial.

How it Works

- Packets are assigned to a priority class.
- Each class has its own queue.
- High-priority queues are processed first.
- Low-priority queues are processed last.
- Provides better QoS than FIFO (less delay for high-priority traffic like multimedia).

Advantages

- Low latency for critical traffic (e.g., VoIP, real-time video).
- Prioritizes latency-sensitive applications.

Disadvantages

- Starvation: If high-priority queues are always full (continuous flow), lower-priority packets never get processed.

Beneficial Example

- VoIP or Real-time Video Conferencing: Prioritized to reduce latency and ensure high-quality communication

R44. What is the single point of failure (SPOF) issue in a hub-and-spoke network design, and how would you mitigate it? / How does dual-homing address SPOF in a hub-and-spoke network, and what are the benefits?

- The Hub is the Single Point of Failure (SPOF).
- If the hub goes down, communication between all spokes is disrupted.
- To prevent this, we could use dual-homing to connect both the hub and spokes to multiple devices or links.
- Dual homing connects devices to two independent devices/links. If one link or device fails, traffic is rerouted through the secondary path.

Benefits of Dual-Homing

- Redundancy: Allows automatic failover in case of failure.
- Improved Availability: If one link fails, the other takes over without significant network downtime.

R2. Hierarchical networks use a tiered design of each layer performing a well defined role in the campus network. Explain the role of each layer in the three-tier architecture. / How does a three-tier system handle redundancy, security and performance? / How does a three-tier architecture handle failover and redundancy?

1. Access Layer (The Network Edge)

- Represents the network edge where traffic enters or exits the campus network.
- An access layer switch provides network access to the user.
- Next-generation switching platforms provide more converged, integrated, and intelligent operations for efficiency and security.

2. Distribution Layer (The Intermediary)

- Interfaces between the Access layer and the Core layer.
- Key Functions:
 - Aggregating Layer 2 broadcast domains and Layer 3 routing.
 - Providing intelligent switching, routing, and network access policy.
 - Providing high availability through redundant distribution layer switches.
 - Providing differentiated services to various classes of service applications at the edge.

3. Core Layer (The Backbone)

- Core Layer, network backbone serves as the aggregator for all distribution layer devices and ties the campus together with the rest of the network.
- To provide fault isolation and high-speed backbone connectivity.

R58. Explain how Software Defined Networking (SDN) differs from traditional networking architectures.

The data plane components of the Traditional approach and the SDN approach identical.

In Traditional Networking,

- The routing algorithm decides the forwarding tables.
- The routing algorithm runs inside every single router.

- Both forwarding and routing happen inside the same router.
- Routers talk to each other to calculate their own forwarding tables.

In SDN (Software-Defined Networking),

- Routing (Control Plane) is separated from the physical router.
- The router only forwards packets.
- A remote controller calculates and sends the forwarding tables.
- The remote controller is in a data center and managed by an ISP or third party.

Tutorial1

1.

R60.Explain the Data Plane (responsible for forwarding packets) and Control Plane (responsible for network decision-making) in SDN.

Data Plane (Forwarding Packets)

- It is responsible for forwarding packets.
- It uses a match-plus-action table. (This replaces the old destination-based table).
- Step 1 (Match): Look up the destination IP address.
- Step 2 (Action): Send the packet to the specified output port.
- Every packet switch has this table. The table is created by a remote controller.

Control Plane (Network Decision-Making)

- It is responsible for making network decisions.
- It is the heart of SDN.
- It calculates the forwarding tables and talks to the routers.
- It is called "software-defined" because this controller is implemented in software, not inside the hardware of each router.

2.

(a)

ANS: Queueing can occur at both the input ports and the output ports of a router.

- Queue occurs on an input port when the arriving rate of packets exceeds the switch capacity.
- Queueing occurs at the output port when datagrams arrive from fabric faster than the transmission rate of output line speed.

(b)

ANS: -HOL blocking a queued packet in an input queue must wait for transfer through the fabric because it is blocked by another packet at the head of the line.

- It occurs at the input port.

Tutorial2

1.

ANS: No, the deny statements will not work as intended.

Key Rules:

- **Top-Down Processing:** ACLs are processed sequentially from the top down. The router stops at the first match and applies that action. It does not read further.
- **Implicit Deny:** Every ACL has an invisible deny any any at the very end. If a packet doesn't match any line, it is dropped.
- **The Problem:** Because permit ip any any is the first line, **every** packet will match it and be permitted. The router will never reach the subsequent deny statements.

How it should be applied:

- Place specific deny statements at the top, followed by permit statements.
- Apply the ACL to an **interface** (using the ip access-group command) and specify the **direction** (in or out) relative to the router.

2.

ANS: For the 50 internal PCs:

Use **Port Address Translation**

- It allows all 50 PCs to share the single public IP address simultaneously. It distinguishes traffic by assigning unique source port numbers to each session.

For the public web server:

Use **Static NAT**.

- The web server must be reachable from the internet at a consistent, unchanging IP address. Static NAT creates a permanent one-to-one mapping between the server's private IP and the public IP.

Inside Local vs. Inside Global Address:

- **Inside Local:** The private IP address assigned to a host inside the company network (e.g., 192.168.1.10). It is not routable on the internet.
- **Inside Global:** The public IP address that represents the inside host to the outside world (e.g., the company's single public IP).

3.

ANS: What is established after Phase 1?

A secure, authenticated **IKE (ISAKMP) SA** (Security Association) is established to allow the peers to negotiate safely.

What must be negotiated in Phase 2?

The **IPsec SA**. This includes the encryption/hashing algorithms and the lifetime for the actual data tunnel.

Two possible causes of communication failure:

1. Mismatched IPsec parameters (e.g., different encryption or hash algorithms in Phase 2).
2. Mismatched Proxy IDs / ACLs (the traffic to be encrypted doesn't match on both sides).

When tunnel mode should be preferred over transport mode

-when connecting two different sites (gateway-to-gateway / Site-to-Site VPN)

Tutorial3

I. A hospital network is being designed to support real-time remote surgery between two hospitals. The application requires a guaranteed amount of bandwidth and very low delay. Before transmission begins, the application wants to request the required network resources. However, if the required resources cannot be reserved along the entire path, the application should not start transmitting. (10 marks)

- a. Which QoS model, IntServ or DiffServ, is more appropriate for this application?
- b. Explain how the selected model can provide the required QoS from the source to the destination.
- c. Identify the signaling protocol and explain the role of admission control in this scenario.
- d. State one major scalability drawback of using this model in a large network.

ANS:

(a) **IntServ (Integrated Services)** is more appropriate.

(b) IntServ provides **hard guarantees** by reserving resources (bandwidth, buffers) for **each individual flow** at **every router** along the path. If all routers successfully reserve the resources, end-to-end QoS is guaranteed.

(c)

- **Signaling Protocol: RSVP.**
- **Role of Admission Control:** Each router checks if it has enough free resources to accept the new request **without affecting existing traffic**. If yes, it reserves them. If no, it rejects the request. This ensures the application **does not start transmitting** unless the whole path can guarantee QoS.

(d) Poor scalability

Each router must store a record for every flow, consuming too much memory and processing power in large networks.

